航动最后一题一限量珍藏版.

$\Omega = 45^\circ$

$\omega = 60^\circ$

$i = 30^\circ$

$e = 0.1$

$a = 20000$

$M_0 = 0$

① 求轨道面参数 i, Ω

任选 $\vec{r_1}$, $\vec{r_2}$

轨外法向 $\vec{N} = \dfrac{\vec{r_1} \times \vec{r_2}}{|\vec{r_1} \times \vec{r_1}|} = \cdots$

$\cos i = \vec{N} \cdot \vec{e_3}$ $\quad i = 30^\circ$

节线 $\vec{ON} = \vec{e_3} \times \vec{N} = (1, 1, 0)$

$\therefore \Omega = 45^\circ$

② 估算动量矩 h

认为 $100s \ll T$. $\vec{v} = \dfrac{\Delta \vec{r}}{\delta t}$

多取几组数据计算 $\left.\begin{matrix} |\vec{r_1} \times \vec{v_1}| \\ |\vec{r_2} \times \vec{v_2}| \\ |\vec{r_3} \times \vec{v_3}| \end{matrix}\right\} \Rightarrow$ 最大的数约为 6.890×10^4

$p = \dfrac{h^2}{\mu} \approx 11868.025\ km$

③ 利用椭圆参数方程 $r = \dfrac{p}{1 + e\cos f}$

$\therefore \Omega = 45°$

② 估算动量矩 h

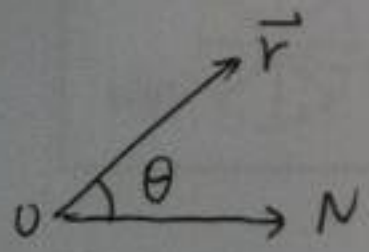
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任意选取两个点 计算出 r_1, r_2.



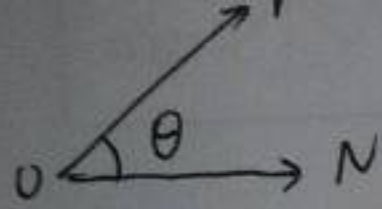
$$\cos\theta = \frac{\vec{r} \cdot \vec{N}}{|\vec{r}|} \Rightarrow \begin{matrix} \cos\theta_1 & \cos\theta_2 \\ \sin\theta_1 & \sin\theta_2 \end{matrix}$$

$$e\cos f = \frac{p}{r} - 1 = e\cos(\theta - \omega)$$

$$= e\left[\cos\theta\cos\omega + \sin\theta\sin\omega\right]$$

$e\cos\omega = x$

$$\begin{cases} x\cos\theta_1 + y\sin\theta_1 = \dfrac{p}{r_1} - 1 \\ \cdots \end{cases} \quad \begin{cases} x = \cdots \\ \cdots \end{cases}$$



$$\frac{\vec{r}}{|\vec{r}|}$$

$\sin\theta_1$ $\sin\theta_2$

$$e\cos f = \frac{p}{r} - 1 = e\cos(\theta - w)$$

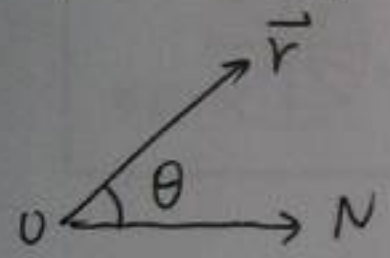
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$$\text{联立}\begin{cases} e\cos w = x \\ e\sin w = y \end{cases} \longrightarrow \begin{cases} x\cos\theta_1 + y\sin\theta_1 = \frac{p}{r_1} - 1 \\ x\cos\theta_2 + y\sin\theta_2 = \frac{p}{r_2} - 1 \end{cases} \Rightarrow \begin{cases} x = \cdots \\ y = \cdots \end{cases}$$

$$\Rightarrow \begin{cases} e = \sqrt{x^2+y^2} = a/ \\ a = \frac{p}{1-e^2} = 20000 \end{cases} \Rightarrow \text{M。你会算哈哈一.}$$

③ 利用椭圆参数方程 $r = \dfrac{p}{1+e\cos f}$

任意选取两个点 计算出 r_1, r_2.



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